DISTRIBUTED NUCLEAR-POWERED LOGISTICS SYSTEM WITH COLLABORATIVE MESH CONTROL

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Abstract

A distributed logistics system comprising a lead power vehicle equipped with a compact nuclear reactor and a plurality of auxiliary transport units mechanically and electrically coupled to form a continuous convoy. The system utilizes a high-voltage power bus and a fiber-optic data bus to distribute energy and control signals from the lead vehicle to the auxiliary units. Each auxiliary unit is equipped with independent electric propulsion and a distributed computing node. A collaborative mesh control algorithm processes real-time data from strain-gauge sensors embedded in inter-vehicle hitches to synchronize motor output, thereby minimizing mechanical tension on the couplers. The system further enables kinetic energy arbitrage, allowing the mass of units on a descent to mechanically assist units on an ascent, and employs a progressive rear-first braking sequence to ensure stability and prevent jackknifing.

Background of the Invention

Long-haul logistics systems, such as freight trains and heavy-truck convoys, are traditionally limited by the tension capacity of mechanical couplers and the fuel requirements of internal combustion engines. In heavy trains, the cumulative draft force on the front couplers often dictates the maximum length and weight of the load, as a single lead engine must pull the entire mass. Furthermore, the need for frequent refueling stops and the environmental impact of fossil fuels present significant operational and ecological challenges. While distributed power in rail (DPUs) and electric trucking are emerging, there lacks a unified system that combines high-density, long-endurance power generation with a cooperative control mesh capable of neutralizing mechanical strain across a massively extended vehicle chain.

Summary of the Invention

The present invention, conceptually referred to as the "Atomic Express," is a distributed, nuclear-powered logistics system designed for ultra-long-haul transport with zero-emission endurance and optimized mechanical stability.

The system comprises a primary power vehicle (the "Lead Vehicle") and a plurality of modular auxiliary transport units ("Nodes"). The Lead Vehicle houses a compact nuclear reactor, preferably a Generation IV Molten Salt Reactor (MSR) or similar Small Modular Reactor (SMR), coupled to a high-output electrical turbine. This rolling power plant generates megawatts of electricity, which is distributed down the length of the convoy via a standardized high-voltage power bus integrated into the inter-vehicle couplers.

Each auxiliary unit is a self-propelled electric vehicle equipped with a distributed computing node. These nodes form a P2P mesh network that communicates via a high-speed fiber-optic bus. The core of the invention is a collaborative control algorithm that treats the convoy as a single, distributed computing system on wheels. By measuring the mechanical strain (tension or compression) at each hitch using embedded piezoelectric sensors, the system adjusts the torque of individual motors in real-time to maintain "Zero-Tension" across the couplers. This ensures that no single coupler is subjected to the cumulative weight of the convoy, effectively removing the physical limits on convoy length.

Furthermore, the system leverages its structural and electrical connectivity to perform kinetic energy arbitrage. On undulating terrain, the gravitational momentum of units on a

downhill slope is used both mechanically (pulling the trailing units) and electrically (via regenerative braking feeding the shared power bus) to assist units currently on an uphill slope. A specialized progressive braking sequence, which activates from the rear units forward, provides a stabilizing "drag-anchor" effect, preventing jackknifing and ensuring safe deceleration of the massively extended convoy.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, world-wide, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the distributed nuclear-powered logistics system, illustrating the lead power vehicle and the chain of auxiliary transport units.
FIG. 2 is a detailed schematic of the inter-vehicle coupling interface, showing the mechanical hitch, electrical power bus, and data link.
FIG. 3 is a flowchart illustrating the collaborative mesh control algorithm for neutralizing coupler strain.
FIG. 4 is a satirical conceptual illustration depicting the original inspiration for the system and establishing a temporal anchor for the conception of the invention.

Detailed Description of the Invention

Referring to **FIG. 1**, the logistics system (100) comprises a lead power vehicle (101) and a plurality of auxiliary units (104, 105, 106). The lead vehicle contains a nuclear reactor core (102) and an electrical generator (103). A high-voltage power bus (107) and a fiber-optic data bus (108) run the length of the convoy. Each auxiliary unit is equipped with an electric motor (109) and a control node.

FIG. 2 illustrates the coupling interface. A mechanical hitch (203) locks the rear of one vehicle (201) to the front of the next (202). Integrated within this hitch is a Megawatt Charging System (MCS) connector (204) for the power bus and a fiber-optic link (205) for the data bus. A piezoelectric strain-gauge sensor (206) provides real-time feedback on the mechanical forces acting on the hitch.

FIG. 3 shows the control logic. The system identifies node mass and position (301), measures real-time hitch strain (302), and checks if the strain exceeds a neutral threshold (303). If so, it adjusts local motor torque (304). This data is synchronized across the mesh (305) to execute cooperative propulsion (306).

FIG. 4 provides the conceptual inspiration for the system, a satirical illustration titled "Atomic Express" which served as the catalyst for the engineering of the distributed, zero-tension logistics model described herein.

Claims

What is claimed is:

1. A distributed logistics system, comprising:
    - a lead power vehicle comprising a nuclear reactor and an electrical generator;
    - a plurality of auxiliary transport units, each comprising an electric propulsion system and a computing node;
    - a coupling system mechanically and electrically linking the lead vehicle and the auxiliary units in a sequence;
    - a shared power bus and a shared data bus extending through the coupling system along the sequence; and
    - a distributed control mesh configured to synchronize the electric propulsion system of each auxiliary unit based on real-time mechanical strain detected at the coupling system.
2. The system of claim 1, wherein the nuclear reactor is a molten salt reactor or a small modular reactor.
3. The system of claim 1, wherein the coupling system comprises piezoelectric sensors configured to measure tension and compression between adjacent units.

4. The system of claim 1, wherein the distributed control mesh is configured to maintain approximately zero tension on the mechanical links of the coupling system by adjusting the torque output of individual auxiliary transport units.

5. The system of claim 1, wherein the system is configured to perform kinetic energy arbitrage by utilizing the gravitational momentum of units on a descent to assist the propulsion of units on an ascent through both mechanical draft and electrical regenerative feedback.

6. The system of claim 1, wherein the system executes a progressive braking sequence starting from the rear-most auxiliary transport unit and moving forward to provide stabilizing drag.

Drawings

FIG. 1: Distributed Nuclear-Powered Logistics System

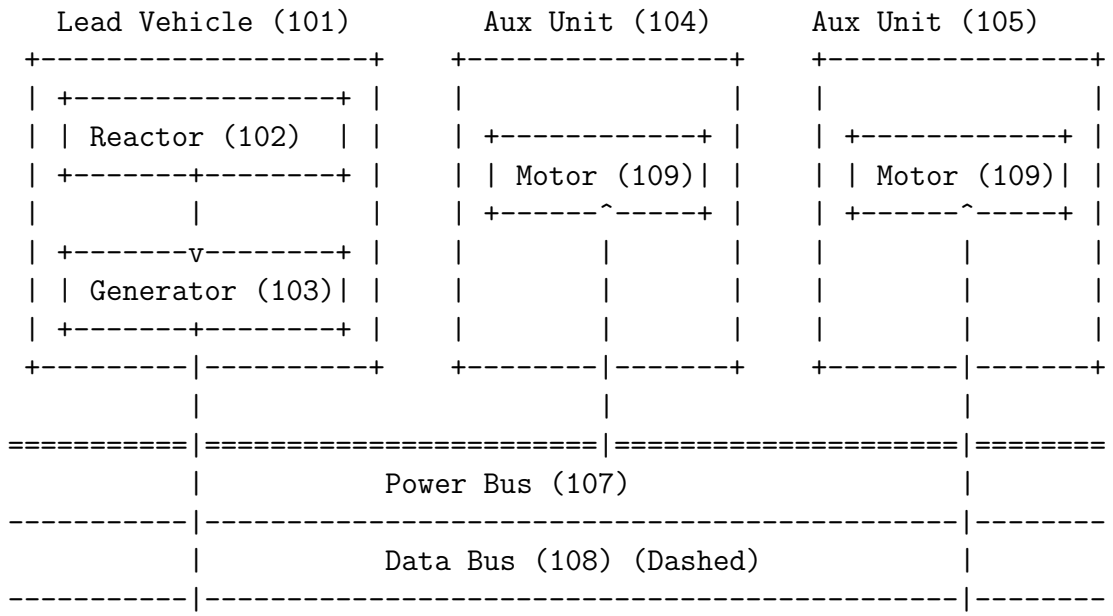


FIG. 2: Inter-Vehicle Coupling Interface

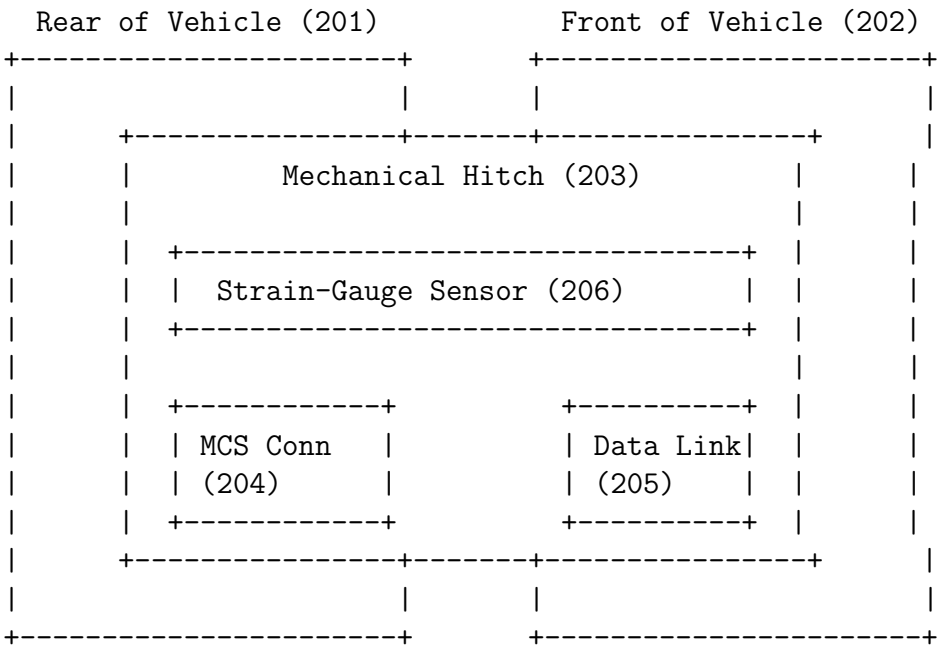


FIG. 3: Distributed Mesh Control Algorithm

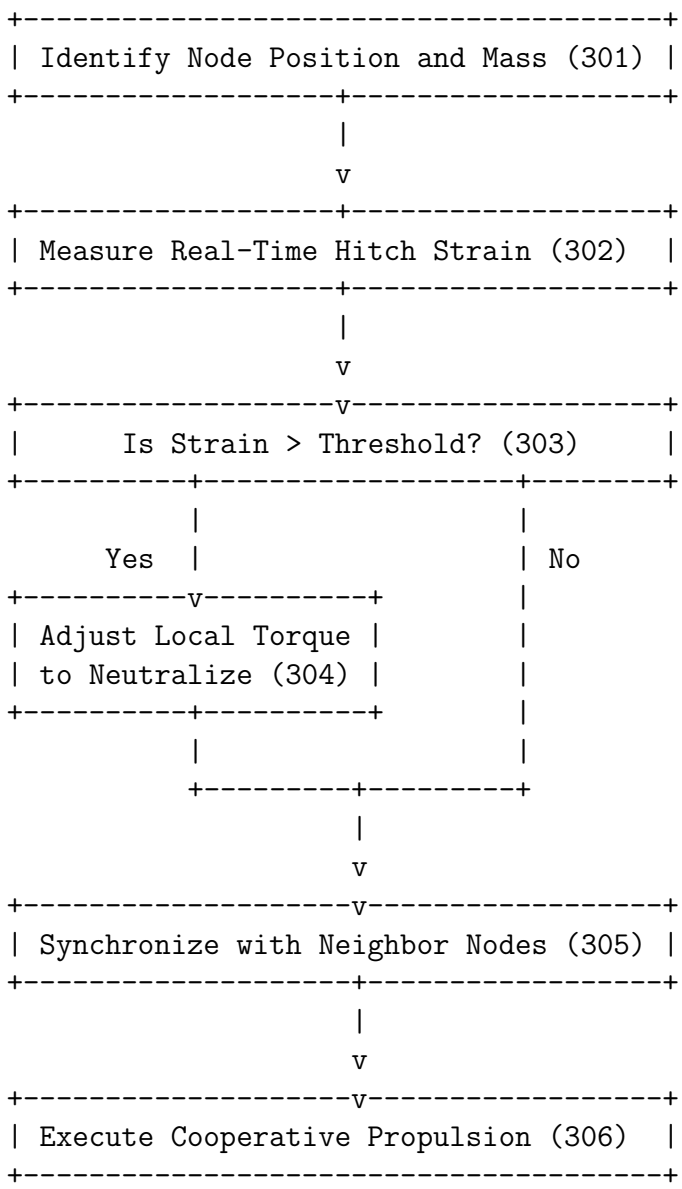




Figure 1: Satirical conceptual illustration of the original inspiration for the system ("Atomic Express").